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# ASSIGNMENT 1

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U26AI103  
NIHIT BHAGWAT  
AI BTECH  
B-33



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Introduction to programming [AI103] Lab

## 1. PROGRAM TO CALCULATE SIMPLE INTEREST

```
1  #include <stdio.h>
2  int main()
3  {
4      float principle, rate, time, intrest;
5      printf("Enter the principle amount: \n");
6      scanf("%f", &principle);
7      printf("Enter the amount of interest: \n");
8      scanf("%f", &rate);
9      printf("Enter the time period: \n");
10     scanf("%f", &time);
11     intrest = (principle * rate * time) / 100;
12     printf("The simple interest is: %f\n", intrest);
13     return 0;
14 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc sim
Enter the principle amount:
25000
Enter the amount of interest:
12
Enter the time period:
6
The simple interest is: 18000.000000
PS C:\VS CODE\cs_assignment_1> █
```

## 2. PROGRAM TO READ MARKS OF FIVE SUBJECT OF A STUDENT AND CALCULATE TOTAL AND PERCENTAGE.

```
1  #include <stdio.h>
2  int main()
3  {
4      float s1, s2, s3, s4, s5, total, percentage;
5      printf("Enter marks of 1st subject : \n");
6      scanf("%f", &s1);
7      printf("Enter marks of 2nd subject : \n");
8      scanf("%f", &s2);
9      printf("Enter marks of 3rd subject : \n");
10     scanf("%f", &s3);
11     printf("Enter marks of 4th subject : \n");
12     scanf("%f", &s4);
13     printf("Enter marks of 5th subject : \n");
14     scanf("%f", &s5);
15     printf("\n\n");
16     if(s1<=100 && s1>=0){
17         printf("s1 : %.2f\n", s1);
18     }
19     else{
20         printf("s1 : Not valid marks");
21     }
22     if(s2<=100 && s2>=0){
23         printf("s2 : %.2f\n", s2);
24     }
25     else{
26         printf("s2 : Not valid marks");
27     }
28     if(s3<=100 && s3>=0){
29         printf("s3 : %.2f\n", s3);
30     }
31     else{
32         printf("s3 : Not valid marks");
33     }
34     if(s4<=100 && s4>=0){
35         printf("s4 : %.2f\n", s4);
36     }
37     else{
38         printf("s4 : Not valid marks");
39     }
40     if(s5<=100 && s5>=0){
41         printf("s5 : %.2f\n", s5);
42     }
43     else{
44         printf("s5 : Not valid marks\n");
45     }
46     total=s1+s2+s3+s4+s5;
47     percentage=(total/5);
48     if((s1>=0 && s1<=100) && (s2>=0 && s2<=100) && (s3>=0 && s3<=100) && (s4>=0 && s4<=100) && (s5>=0 && s5<=100) ){
49         printf("Total marks is : %.2f\n", total);
50         printf("Total percentage is : %.2f\n", percentage);
51     }
52     else{
53         printf("Total marks : ERROR\n");
54         printf("Total percentage : ERROR\n");
55     }
56     return 0;
57 }
```

OUTPUT:

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\lab_1"
Enter marks of 1st subject :
96
Enter marks of 2nd subject :
98
Enter marks of 3rd subject :
95
Enter marks of 4th subject :
63
Enter marks of 5th subject :
91

s1 : 96.00
s2 : 98.00
s3 : 95.00
s4 : 63.00
s5 : 91.00
Total marks is : 443.00
Total percentage is : 88.60
PS C:\VS CODE\cs_assignment_1\lab_1> █
```

### 3. PROGRAM TO CALCULATE GROSS SALARY.

```
1  #include<stdio.h>
2  int main()
3  {
4      float salary, CTC, employee_benefits;
5      printf("Enter your CTC salary: \n");
6      scanf("%f", &CTC);
7      printf("Enter your employee benefits: \n");
8      scanf("%f", &employee_benefits);
9      salary = CTC - employee_benefits;
10     printf("your gross salary is: %f", salary);
11 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc pr
e_gross_salary }
Enter your CTC salary:
120000
Enter your employee benefits:
20000
your gross salary is: 100000.000000
PS C:\VS CODE\cs_assignment_1> 
```

#### 4. PROGRAM TO CONVERT TEMPRATURE FROM FAHRENHEIT TO CENTIGRADE DEGREES.

```
1  #include<stdio.h>
2  int main()
3  {
4      float fahrenheit, celsius;
5      printf("Enter the temprature in fahrenheit: \n");
6      scanf("%f", &fahrenheit);
7      celsius = (fahrenheit - 32)*5/9;
8      printf("The temprature in celsius is = %f", celsius);
9      return 0;
10 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc fahrenheit.c
Enter the temprature in fahrenheit:
110
The temprature in celsius is = 43.333332
PS C:\VS CODE\cs_assignment_1> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc fahrenheit.c
Enter the temprature in fahrenheit:
120
The temprature in celsius is = 48.888889
PS C:\VS CODE\cs_assignment_1> █
```

## 5. PROGRAM TO SWAP TWO VARIABLES USING THIRD NUMBER.

```
1  #include <stdio.h>
2  int main()
3  {
4      int first, second, third;
5      printf("Enter two numbers : ");
6      scanf("%d %d", &first, &second);
7      //now swap two numbers by third//
8      third = first;
9      first=second;
10     second=third;
11     printf("After swappping the numbers we get : %d %d", first, second);
12 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc swapping_by_third_number.
Enter two numbers : 5
6
After swappping the numbers we get : 6 5
PS C:\VS CODE\cs_assignment_1> |
```

6. PROGRAM TO SWAP TWO VARIABLES WITHOUT USING THIRD VARIABLES.

```
1  #include <stdio.h>
2  int main()
3  {
4      int a, b;
5      printf("Enter two numbers: \n");
6      scanf("%d %d", &a, &b);
7      a=a+b;
8      b=a-b;
9      a=(a-b);
10     printf("After swapping the numbers : %d %d", a, b);
11     return 0;
12 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc swappin
er }
Enter two numbers:
9
19
After swapping the numbers : 19 9
PS C:\VS CODE\cs_assignment_1> 
```



## 7. PROGRAM TO CALCULATE AREA OF TRIANGLE

```
cs_assignment_1 > area_of_triangle.c > main()
1  #include<stdio.h>
2  #include<math.h>
3  int main()
4  {
5      float a,b,c,s,area;
6      printf("Enter three sides of a triangle: \n");
7      scanf("%f %f %f", &a, &b, &c);
8      s=(a+b+c)/2;
9      area=sqrt(s*(s-a)*(s-b)*(s-c));
10     printf("Area of the triangle is = %f", area);
11     return 0;
12 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc a
Enter three sides of a triangle:
5
6
7
Area of the triangle is = 14.696939
PS C:\VS CODE\cs_assignment_1> 
```

8. EARTH TAKES A PERIOD OF REVOLUTION OF 31558150 SECONDS. WRITE A C PROGRAM TO CONVERT THIS INTO NUMBER OF DAY, HOURS AND MINUTES

```
cs_assignment_1 / ... earth.c ...
1  #include<stdio.h>
2  int main()
3  {
4      float seconds, days, hours, minutes;
5      seconds = 31558150;
6      minutes = seconds/60;
7      hours = minutes/60;
8      days = hours/24;
9      printf("Total number of days in a year is: %f\n", days);
10     printf("total no. of hours in a years is: %f\n", hours);
11     printf("total no. of minutes in a year is : %f\n", minutes);
12     return 0;
13 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1" ; if ($?) { gcc earth.c -o earth } ; if ($?) { .\earth }
Total number of days in a year is: 365.256378
total no. of hours in a years is: 8766.153320
total no. of minutes in a year is : 525969.187500
PS C:\VS CODE\cs_assignment_1> |
```

9. PROGRAM TO READ TIME IN HR, MIN, SEC AND CONVERT IT INTO TOTAL SECOND.

```
1  #include <stdio.h>
2  int main()
3  {
4      float time, hours, minutes, seconds, total_seconds;
5      printf("Enter hours:\n");
6      scanf("%f", &hours);
7      printf("Enter minutes:\n");
8      scanf("%f", &minutes);
9      printf("Enter seconds:\n");
10     scanf("%f", &seconds);
11     total_seconds = ((hours*60)*60)+(minutes*60)+seconds;
12     printf("Total time in seconds is : %f", total_seconds);
13     return 0;
14 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc 9_time.c -o 9_
Enter hours:
8
Enter minutes:
35
Enter seconds:
53
Total time in seconds is : 30953.000000
PS C:\VS CODE\cs_assignment_1> |
```

**10.** WRITE A C PROGRAM TO CALCULATE THE CUT OFF MARK OF A STUDENT USING THE FORMULA.

$$CM = M/2 + P/2 + C/2 + E$$

WHERE CM = Cut off mark

M = Marks in Mathematics out of 200

P = Marks in Physics out of 200

C = Marks in Chemistry out of 200

E = Marks in entrance examination out of 100.

```
1  #include <stdio.h>
2  int main()
3  {
4      float M, P, C, E, total_marks, CM;
5      printf("Enter marks of maths :\n");
6      scanf("%f", &M);
7      printf("Enter marks of physics :\n");
8      scanf("%f", &P);
9      printf("Enter marks of chemistry :\n");
10     scanf("%f", &C);
11     printf("Enter marks of Entrance exam :\n");
12     scanf("%f", &E);
13     if(M<=200)
14         printf("Marks obtained in Maths is : %f\n", M);
15     else
16         printf("Not valid\n");
17     if(P<=200)
18         printf("Marks obtained in Physics is : %f\n", P);
19     else
20         printf("Not valid\n");
21     if(C<=200)
22         printf("Marks obtained in Chemistry is : %f\n", C);
23     else
24         printf("Not valid\n");
25     if(E<=100)
26         printf("Marks obtained in Entrance exam is : %f\n", E);
27     else
28         printf("Not valid\n");
29     CM = M/2 + P/2 + C/2 + E;
30     if(M<=200 && P<=200 && C<=200 && E<=100)
31         printf("Cut off marks of student is : %f", CM);
32     else
33         printf("CUT OFF MARKS : INVALID\n");
34     return 0;
35 }
```

OUTPUT:

```
1  #include <stdio.h>
2  int main()
3  {
4      float M, P, C, E, total_marks, CM;
5      printf("Enter marks of maths :\n");
6      scanf("%f", &M);
7      printf("Enter marks of physics :\n");
8      scanf("%f", &P);
9      printf("Enter marks of chemistry :\n");
10     scanf("%f", &C);
11     printf("Enter marks of Entrance exam :\n");
12     scanf("%f", &E);
13     if(M<=200)
14         printf("Marks obtained in Maths is : %f\n", M);
15     else
16         printf("Not valid\n");
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc 10_
Enter marks of maths :
196
Enter marks of physics :
189
Enter marks of chemistry :
173
Enter marks of Entrance exam :
86
Marks obtained in Maths is : 196.000000
Marks obtained in Physics is : 189.000000
Marks obtained in Chemistry is : 173.000000
Marks obtained in Entrance exam is : 86.000000
Cut off marks of student is : 365.000000
PS C:\VS CODE\cs_assignment_1> █
```

## 11. PROGRAM TO READ TOTAL SECOND AND CONVERT IT INTO TIME.

```
1  #include<stdio.h>
2  int main()
3  {
4      int total_seconds, hours, minutes, seconds;
5      printf("Enter total seconds: \n");
6      scanf("%d", &total_seconds);
7      hours = total_seconds/3600;
8      minutes = total_seconds%3600/60;
9      seconds = total_seconds-(hours*3600 + minutes*60);
10     printf("Time is : \n");
11     printf("%d hours %d minutes %d seconds", hours, minutes, seconds);
12     return 0;
13 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc 11_second_to_time.c
Enter total seconds:
19836
Time is :
5 hours 30 minutes 36 seconds
PS C:\VS CODE\cs_assignment_1> |
```

## 12. PROGRAM TO PRINT ASCII CODE OF ANY CHARACTER.

cs\_assignment\_1 > C ASCII\_CODE.c > main()

```
1  #include <stdio.h>
2  int main()
3  {
4      char ch;
5      printf("Enter any character\n");
6      scanf("%c", &ch);
7      printf("ASCII value of %c is %d", ch, ch);
8      return 0;
9  }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc ASCII_CODE.c -o ASCII_CODE } ; if ($?) { .\ASCII_CODE }
Enter any character
k
ASCII value of k is 107
PS C:\VS CODE\cs_assignment_1> |
```

### 13. PROGRAM TO CHECK WHETHER A NO. IS EVEN OR ODD.

```
cs_assignment_1 > C even_or_odd.c > ...
1  #include <stdio.h>
2  int main()
3  {
4      int num;
5      printf("Enter any number\n");
6      scanf("%d", &num);
7      if (num % 2 == 0)
8      {
9          printf("%d is an even number", num);
10     }
11     else
12     printf("%d is an odd number", num);
13     return 0;
14 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) {
Enter any number
6
6 is an even number
PS C:\VS CODE\cs_assignment_1>
PS C:\VS CODE\cs_assignment_1> cd "c:\VS CODE\cs_assignment_
Enter any number
9
9 is an odd number
PS C:\VS CODE\cs_assignment_1> |
```



**14. PROGRAM TO CHECK WHETHER A GIVEN CHARACTER IS CAPITAL, LETTER, SMALL CASE LETTER, A DIGIT OR A SPECIAL SYMBOL.**

```
cs_assignment_1 > C 14_IDENTIFIER.c > main()
1  #include <stdio.h>
2  int main()
3  {
4      char ch;
5      printf("Enter any character :");
6      scanf("%c", &ch);
7      //check character type using ASCII values
8      if (ch >= 'A' && ch <= 'Z')
9      {
10         printf("The character %c is a CAPITAL LETTER.\n", ch);
11     }
12     else if (ch >= 'a' && ch <= 'z')
13     {
14         printf("The character %c is a small case letter.\n", ch);
15     }
16     else if (ch >= '0' && ch <= '9')
17     {
18         printf("The character %c is a digit.\n", ch);
19     }
20     else
21     {
22         printf("The character %c is a special character.\n", ch);
23     }
24     return 0;
25 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc 14_IDENTIFIER.c
Enter any character :l
The character l is a small case letter.
PS C:\VS CODE\cs_assignment_1> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) { gcc
Enter any character :*
The character * is a special character.
PS C:\VS CODE\cs_assignment_1> █
```

15. PROGRAM TO READ MARKS OF FIVE SUBJECT AND PRINT DIVISION.

cs\_assignment\_1 > C 15.c > ...

```
1  #include <stdio.h>
2  int main()
3  {
4      float s1, s2, s3, s4, s5, total, percentage;
5      printf("Enter the marks of 5 subjects: \n");
6      scanf("%f %f %f %f %f", &s1, &s2, &s3, &s4, &s5);
7      total=s1+s2+s3+s4+s5;
8      percentage=total/5;
9      printf("Total marks=%.2f\n", total);
10     printf("Total percentage=%.2f\n", percentage);
11     if(percentage>=70)
12     {
13         printf("GRADE A");
14     }
15     else if(percentage>=50 && percentage < 70)
16     {
17         printf("GRADE B");
18     }
19     else if(percentage>=35 && percentage<50)
20     {
21         printf("GRADE C");
22     }
23     else
24     {
25         printf("FAIL");
26     }
27     return 0;
28 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\cs_assignment_1\" ; if ($?) {
Enter the marks of 5 subjects:
56
23
98
84
96
Total marks=357.00
Total percentage=71.40
GRADE A
```